Online Appendix C to "Assortment Optimization with Customer-Centric Fairness"

Appendix C.1. Proofs of Section 5

Proof of Lemma 6: First, for any pair in the support of an optimal policy, write $C_i = |\mathcal{S}_i|$ and replace $\mathcal{S}_i$ by $\mathcal{S}_i^*(\widehat{C}_i)$, where $\widehat{C}_i = \min\{|\mathcal{S}_2^D|, \max\{|\mathcal{S}_1^D|, C_i\}\}$, $i = 1, 2$. Lemma 2 implies that this replacement does not decrease revenue, and $(\widehat{C}_i - \widehat{C}_j - \alpha_{i,j})^+ \le (C_i - C_j - \alpha_{i,j})^+$, $i, j \in \{1, 2\}$, so both fairness constraints remain satisfied. Thus, we can restrict all support pairs to $|\mathcal{S}_1^D| \le |\mathcal{S}_i| \le |\mathcal{S}_2^D|$, $i = 1, 2$, without losing optimality.

Assume that there exists an optimal strategy G_0 such that $P_{G_0}((\mathcal{S}_1^0, \mathcal{S}_2^0)) > 0$ and $|\mathcal{S}_1^0| > |\mathcal{S}_2^0|$. Let $\boldsymbol{\mathcal{S}}^0 = (\mathcal{S}_1^0, \mathcal{S}_2^0)$ and $\widehat{\boldsymbol{S}} = (\mathcal{S}_1^0, \mathcal{S}_2^*(|\mathcal{S}_1^0|))$. These pairs are distinct because their second assortments have different cardinalities. Construct G' by setting

$$P_{G'}(\boldsymbol{S}) = \begin{cases} 0, & \boldsymbol{S} = \boldsymbol{\mathcal{S}}^0, \\ P_{G_0}(\widehat{\boldsymbol{S}}) + P_{G_0}(\boldsymbol{\mathcal{S}}^0), & \boldsymbol{S} = \widehat{\boldsymbol{S}}, \\ P_{G_0}(\boldsymbol{S}), & \text{otherwise.} \end{cases}$$

Here, $\mathcal{S}_2^*(|\mathcal{S}_1^0|)$ is an optimal assortment for group 2 with cardinality $|\mathcal{S}_1^0|$. We will show G' is feasible and will not decrease the revenue. First, the probability is well-defined and satisfies the fairness constraints since $\mathbb{E}_{G'}\left[(|\mathcal{S}_2| - |\mathcal{S}_1| - \alpha_{2,1})^+\right] = \mathbb{E}_{G_0}\left[(|\mathcal{S}_2| - |\mathcal{S}_1| - \alpha_{2,1})^+\right] \le \kappa_1$ and $\mathbb{E}_{G'}\left[(|\mathcal{S}_1| - |\mathcal{S}_2| - \alpha_{1,2})^+\right] \le \mathbb{E}_{G_0}\left[(|\mathcal{S}_1| - |\mathcal{S}_2| - \alpha_{1,2})^+\right] \le \kappa_2$. Second, by the initial cardinality restriction, we have $|\mathcal{S}_1^0| \le |\mathcal{S}_2^D|$ so the revenue for group 2 will not decrease when increasing the cardinality of $\mathcal{S}_2$.

We can repeat the process until we construct a strategy such that $P_G((\mathcal{S}_1, \mathcal{S}_2)) = 0$ for every $|\mathcal{S}_1| > |\mathcal{S}_2|$, and the revenue is no less than the maximal revenue brought by G_0. Then we finish the proof. □

Proof of Theorem 4: It is obvious that if the fully customized assortments satisfy $|\mathcal{S}_2^D| - |\mathcal{S}_1^D| - \alpha_{2,1} \le \kappa_1$, then an optimal strategy G^F is deterministic and $P_{G^F}((\mathcal{S}_1^D, \mathcal{S}_2^D)) = 1$. If this condition fails and $\kappa_1 = 0$, Lemma 6 allows us to restrict attention to pairs with $|\mathcal{S}_1| \le |\mathcal{S}_2|$. Every pair with positive probability then satisfies $|\mathcal{S}_2| - |\mathcal{S}_1| \le \alpha_{2,1}$, so an optimal deterministic fair pair is optimal for the randomized problem. It remains to consider $0 < \kappa_1 < |\mathcal{S}_2^D| - |\mathcal{S}_1^D| - \alpha_{2,1}$. Based on Lemma 6, $\mathbb{E}_{G^F}\left[(|\mathcal{S}_1| - |\mathcal{S}_2| - \alpha_{1,2})^+\right] = 0$ and we only need to consider the probability on the set $\mathcal{W} = \left\{(\mathcal{S}_1, \mathcal{S}_2) : |\mathcal{S}_1^D| \le |\mathcal{S}_1| \le |\mathcal{S}_2| \le |\mathcal{S}_2^D|\right\}$. Then Problem (RFA-2) can be rewritten as

$$\begin{aligned} &\max_{P_G} \sum_{(\mathcal{S}_1, \mathcal{S}_2) \in \mathcal{W}} P_G((\mathcal{S}_1, \mathcal{S}_2)) \left(\sum_{i=1}^{2} \lambda_i R_i(\mathcal{S}_i) \right), \\ &\text{s.t.} \sum_{(\mathcal{S}_1, \mathcal{S}_2) \in \mathcal{W}} P_G((\mathcal{S}_1, \mathcal{S}_2))(|\mathcal{S}_2| - |\mathcal{S}_1| - \alpha_{2,1})^+ = \kappa_1, \quad \sum_{(\mathcal{S}_1, \mathcal{S}_2) \in \mathcal{W}} P_G((\mathcal{S}_1, \mathcal{S}_2)) = 1, \quad P_G((\mathcal{S}_1, \mathcal{S}_2)) \ge 0, \forall (\mathcal{S}_1, \mathcal{S}_2) \in \mathcal{W}, \end{aligned} \tag{A.8}$$

The equality in the first constraint can be attained by an optimal policy: if it is strict, mix that policy with $(\mathcal{S}_1^D, \mathcal{S}_2^D)$ until equality holds. Since the fully customized pair maximizes revenue and has excess gap greater than κ_1, this preserves feasibility without decreasing revenue.

Note that for each $(\mathcal{S}_1, \mathcal{S}_2)$, the corresponding revenue $\sum_{i=1}^{2} \lambda_i R_i(\mathcal{S}_i)$ and the excess cardinality difference $(|\mathcal{S}_2| - |\mathcal{S}_1| - \alpha_{2,1})^+$ are constants. Therefore, Problem (A.8) is a linear programming problem in terms of $P_G((\mathcal{S}_1, \mathcal{S}_2))$, $(\mathcal{S}_1, \mathcal{S}_2) \in \mathcal{W}$ with two constraints. Based on the standard result of the linear programming problem, there exists an optimal strategy G^F with at most two decision variables taking positive values.

Therefore, we assume there exists an optimal randomized strategy G^F that is supported on at most two assortment pairs $(\mathcal{S}_1^1, \mathcal{S}_2^1)$ and $(\mathcal{S}_1^2, \mathcal{S}_2^2)$, with the cardinality difference between the two groups $C_j = |\mathcal{S}_2^j| - |\mathcal{S}_1^j|$, $j = 1, 2$ and the corresponding probability $P_j = P_G((\mathcal{S}_1^j, \mathcal{S}_2^j))$, $j = 1, 2$. Assume $C_1 \le C_2$ without loss of generality. We can also denote $R(C) = \max_{|\mathcal{S}_2|-|\mathcal{S}_1|=C}\{\sum_{i=1}^2 \lambda_i R_i^*(|\mathcal{S}_i|)\}$, where the maximum is over $(\mathcal{S}_1, \mathcal{S}_2) \in \mathcal{W}$ and $R_i^*(|\mathcal{S}_i|)$ is the maximal revenue with cardinality $|\mathcal{S}_i|$. Then the problem is equivalent to

$$\begin{aligned} \max_{C_1, C_2 \in \{0,\dots,|\mathcal{S}_2^D|-|\mathcal{S}_1^D|\}} \max_{P_1, P_2 \ge 0} & P_1 R(C_1) + P_2 R(C_2), \\ \text{s.t. } & P_1(C_1 - \alpha_{2,1})^+ + P_2(C_2 - \alpha_{2,1})^+ = \kappa_1, P_1 + P_2 = 1. \end{aligned} \tag{A.9}$$

If only one assortment pair has positive probability, denote it by $(\mathcal{S}_1^1, \mathcal{S}_2^1)$ and set $P_1 = 1$. The first constraint above and $\kappa_1 > 0$ then imply $C_1 = \alpha_{2,1} + \kappa_1$. We choose the fully customized pair $(\mathcal{S}_1^D, \mathcal{S}_2^D)$ as $(\mathcal{S}_1^2, \mathcal{S}_2^2)$ and set $P_2 = 0$. Since $\kappa_1 < |\mathcal{S}_2^D| - |\mathcal{S}_1^D| - \alpha_{2,1}$, we have

$$C_1 = \alpha_{2,1} + \kappa_1 < |\mathcal{S}_2^D| - |\mathcal{S}_1^D| = C_2.$$

If both assortment pairs have positive probabilities and $C_1 = C_2$, we assign probability one to the pair with the larger revenue. This replacement leaves the fairness constraint unchanged and does not decrease revenue. The resulting policy has only one assortment pair with positive probability, so the preceding construction applies. Therefore, we may assume $C_1 < C_2$.

Then we can solve the problem by discussing the value of C_j.

(i) When $C_1 \le C_2 \le \alpha_{2,1}$, we have $0P_1 + 0P_2 = 0 < \kappa_1$, so the equality constraint cannot hold.

(ii) When $C_1 \le \alpha_{2,1} \le C_2$, the constraints become $P_2(C_2 - \alpha_{2,1}) = \kappa_1$ and $P_1 + P_2 = 1$, which are independent with C_1. Note that $|\mathcal{S}_2^D| - |\mathcal{S}_1^D| - C_1 \ge |\mathcal{S}_2^D| - |\mathcal{S}_1^D| - \alpha_{2,1} > \kappa_1 > 0$, revealing the fully customized assortment is unfair with tolerance level C_1. Then based on Theorem 1, the optimal objective value of Problem (DFA-2) with tolerance levels $(\alpha_{1,2}, C_1)$ is equal to $R(C_1)$ as we defined above, so $R(C_1)$ increases in C_1. In combination with the fact that C_1 does not change the solution space, the objective value of Problem (A.9) increases in C_1 and thus we can choose $C_1 = \alpha_{2,1}$ without decreasing revenue. Under the case $C_1 = \alpha_{2,1}$, this problem can be regarded as a special case of Case (iii) below.

(iii) When $\alpha_{2,1} \le C_1 < C_2$, the constraints become $P_1(C_1 - \alpha_{2,1}) + P_2(C_2 - \alpha_{2,1}) = \kappa_1$ and $P_1 + P_2 = 1$, and we solve the inner linear programming problem with the optimal solution to be $P_1 = \frac{C_2 - \alpha_{2,1} - \kappa_1}{C_2 - C_1}$, $P_2 = \frac{\alpha_{2,1} + \kappa_1 - C_1}{C_2 - C_1}$. These probabilities are well-defined because $C_1 < C_2$, and they are nonnegative exactly when $\alpha_{2,1} \le C_1 \le \alpha_{2,1} + \kappa_1 \le C_2$. Therefore, the optimal solution can be represented in the form:

$$P_{G^F}((\mathcal{S}_1, \mathcal{S}_2)) = \begin{cases} \frac{|\mathcal{S}_2^2| - |\mathcal{S}_1^2| - \alpha_{2,1} - \kappa_1}{|\mathcal{S}_2^2| - |\mathcal{S}_1^2| - |\mathcal{S}_2^1| + |\mathcal{S}_1^1|} & \text{if } (\mathcal{S}_1, \mathcal{S}_2) = (\mathcal{S}_1^1, \mathcal{S}_2^1), \\ \frac{\alpha_{2,1} + \kappa_1 - |\mathcal{S}_2^1| + |\mathcal{S}_1^1|}{|\mathcal{S}_2^2| - |\mathcal{S}_1^2| - |\mathcal{S}_2^1| + |\mathcal{S}_1^1|} & \text{if } (\mathcal{S}_1, \mathcal{S}_2) = (\mathcal{S}_1^2, \mathcal{S}_2^2), \\ 0 & \text{otherwise,} \end{cases}$$

where $|\mathcal{S}_2^2| - |\mathcal{S}_1^2| \ge \alpha_{2,1} + \kappa_1 \ge |\mathcal{S}_2^1| - |\mathcal{S}_1^1| \ge \alpha_{2,1}$. The result $|\mathcal{S}_1^D| \le |\mathcal{S}_i^j| \le |\mathcal{S}_2^D|$, $i, j \in \{1, 2\}$ follows from the cardinality restriction established in the proof of Lemma 6. The construction above also ensures $|\mathcal{S}_2^1| - |\mathcal{S}_1^1| < |\mathcal{S}_2^2| - |\mathcal{S}_1^2|$, including when one probability is zero. This completes the proof. □

Proof of Theorem 5: We first prove that Algorithm A.4 can achieve the optimal solution. The two early returns in Step 1 are optimal by Theorem 4. We therefore consider $0 < \kappa_1 < |\mathcal{S}_2^D| - |\mathcal{S}_1^D| - \alpha_{2,1}$. In Step 2, we know that $R_i^*(C_i)$, $i = 1, 2$, gradually decreases after each iteration based on Lemma 2. Therefore, the position of $R_i^*(C_i)$, that is, $R_i^*(C) \in [I_i^{(\bar{z})}, I_i^{(\bar{z}+1)})$ is also decreasing after each iteration. Like the proof of Algorithm A.1, Lemma B.3 shows that $\bar{z}_i$ in Step 2.1 successfully records the position of $R_i^*(C_i)$. In this case, $\mathcal{S}_i(C_i) = \mathcal{B}_i^{(\bar{z}_i)}(C_i)$, $i = 1, 2$ is the optimal assortment

for group i with cardinality C_i. In Step 3, we denote the cardinality of group 1 to be C and that of group 2 to be $C+\Delta$. Recall that $R(\Delta)$ is the maximal total revenue at cardinality gap Δ. It can be calculated by

$$\begin{aligned} R(\Delta) &= \max_{C\in\mathcal{Z}^+\cap[|\mathcal{S}_1^D|,|\mathcal{S}_2^D|-\Delta]} \left(\lambda_1 R_1^*(C)+\lambda_2 R_2^*(C+\Delta)\right) \\ &= \max_{C\in\mathcal{Z}^+\cap[|\mathcal{S}_1^D|,|\mathcal{S}_2^D|-\Delta]} \left(\lambda_1 R_1(\mathcal{S}_1(C))+\lambda_2 R_2(\mathcal{S}_2(C+\Delta))\right), \end{aligned}$$

where $\mathcal{S}_i(C_i)$ is defined as above. Therefore, Step 3 can provide the value of $R(\Delta)$ with $\Delta \in \{\alpha_{2,1},\dots,|\mathcal{S}_2^D|-|\mathcal{S}_1^D|\}$. Finally, in Step 4, we can calculate the optimal revenue of Problem (RFA-2) based on Theorem 4:

$$\begin{aligned} \Pi &= \max_{\mathcal{S}_i^j,i,j=1,2} \tfrac{|\mathcal{S}_2^2|-|\mathcal{S}_1^2|-\alpha_{2,1}-\kappa_1}{|\mathcal{S}_2^2|-|\mathcal{S}_1^2|-|\mathcal{S}_2^1|+|\mathcal{S}_1^1|} R(|\mathcal{S}_2^1|-|\mathcal{S}_1^1|) + \tfrac{\alpha_{2,1}+\kappa_1-|\mathcal{S}_2^1|+|\mathcal{S}_1^1|}{|\mathcal{S}_2^2|-|\mathcal{S}_1^2|-|\mathcal{S}_2^1|+|\mathcal{S}_1^1|} R(|\mathcal{S}_2^2|-|\mathcal{S}_1^2|) \\ &= \max_{\Delta_1,\Delta_2} \tfrac{\Delta_2-\alpha_{2,1}-\kappa_1}{\Delta_2-\Delta_1} R(\Delta_1) + \tfrac{\alpha_{2,1}+\kappa_1-\Delta_1}{\Delta_2-\Delta_1} R(\Delta_2), \text{s.t. } \alpha_{2,1}\le\Delta_1\le\alpha_{2,1}+\kappa_1\le\Delta_2, \end{aligned}$$

where $\Delta_j = |\mathcal{S}_2^j| - |\mathcal{S}_1^j|$, $j=1,2$,and both maximizations are restricted to $\Delta_1,\Delta_2 \in \{\alpha_{2,1},\dots,|\mathcal{S}_2^D|-|\mathcal{S}_1^D|\}$ with $\Delta_1<\Delta_2$. If the optimal policy characterized in Theorem 4 has only one assortment pair with positive probability, this pair satisfies $\Delta_1=\alpha_{2,1}+\kappa_1$. Choose $\Delta_2 = |\mathcal{S}_2^D|-|\mathcal{S}_1^D| > \Delta_1$. The two coefficients in the objective of Step 4 then satisfy $\frac{\Delta_2-\alpha_{2,1}-\kappa_1}{\Delta_2-\Delta_1}=1$, $\frac{\alpha_{2,1}+\kappa_1-\Delta_1}{\Delta_2-\Delta_1}=0$. Thus, Step 4 also includes this policy by assigning probability one to the first assortment pair and zero to the second. Then we finish the proof.

Then we analyze the complexity of the algorithm. The early returns require at most $O(N^2\log N)$ time by Theorem 2. Given the interval information, the time complexity of Step 2 is the same as that of Step 2.1 in Algorithm A.1, which is $O(N^2)$. The complexity of Steps 3 and 4 is also $O(N^2)$. The interval construction in Step 1, as described in Appendix A.1, adds $O(N^2\log N)$ time, yielding the same overall bound as in Theorem 2. □

Proof of Proposition 1. We establish the cardinality-based reformulation, the restricted cardinality range, and the support-size bound.

(i) Fixed-cardinality reformulation. First, use the same definitions of $F(\boldsymbol{C})$ and $U_i(\boldsymbol{C})$ for all $\boldsymbol{C}\in\{0,1,\dots,N\}^M$. Given any feasible randomized assortment policy, replace each offered assortment $\mathcal{S}_i$ by $\mathcal{S}_i^*(|\mathcal{S}_i|)$. This replacement leaves every cardinality, and hence every unfairness term, unchanged. It also weakly increases revenue because $R_i(\mathcal{S}_i)\le R_i^*(|\mathcal{S}_i|)$. Aggregate the probabilities of all joint assortments with the same cardinality vector into $p_{\boldsymbol{C}}$. The resulting policy has expected revenue $\sum_{\boldsymbol{C}} p_{\boldsymbol{C}} F(\boldsymbol{C})$, expected unfairness $\sum_{\boldsymbol{C}} p_{\boldsymbol{C}} U_i(\boldsymbol{C})$ for type i, and total probability one. Thus, before restricting the cardinality range, the assortment problem can be converted into its cardinality-based linear program.

(ii) Restriction to $\mathcal{D}^M$. Consider any cardinality vector C in the support of an optimal randomized policy. Suppose $\min_i C_i < D_1$, and let $I=\{i: C_i=\min_m C_m\}$. Construct a new vector $\widetilde{C}$ by increasing C_i by one for all $i\in I$, while keeping all other components unchanged. For every $i\in I$, we have $C_i < D_1 \le D_i$, so $R_i^*(C_i+1)\ge R_i^*(C_i)$ by the unimodality of $R_i^*(\cdot)$. Hence the revenue does not decrease.

Moreover, no unfairness term increases. Increasing the smallest components reduces the unfairness perceived by these types, and it does not create new unfairness for other types because the increased components remain no larger than the other components. Therefore, replacing C by $\widetilde{C}$ preserves feasibility and weakly improves revenue. Repeating this operation eliminates all support vectors with $\min_i C_i < D_1$.

Similarly, suppose $\max_i C_i > D_M$, and let $I=\{i: C_i=\max_m C_m\}$. Decreasing all components in I by one weakly improves revenue because $C_i > D_M \ge D_i$ for all $i\in I$, and again no unfairness term increases. Repeating this operation eliminates all support vectors with $\max_i C_i > D_M$. Thus there exists an optimal randomized policy supported only on cardinality vectors satisfying $D_1\le C_i\le D_M$ for every i.

Overall, Part (i) together with the restriction in Part (ii) show that Problem (RFA-M) is exactly equivalent to Problem (9).

(iii) Support-size bound. The column of the coefficient vector in the linear programming problem (9) is $\big(1, U_1(C), \ldots, U_M(C)\big)^\top \in \mathbb{R}^{M+1}$. Therefore, at most $M+1$ among such vectors are linearly independent. Here, adding one slack variable for each fairness inequality gives a standard-form linear program with $M+1$ equality constraints. Its feasible set is nonempty, and the objective is bounded. An optimal basic solution therefore exists. Based on Theorem 2.4 in Bertsimas and Tsitsiklis (1997), there exists an optimal basic solution that has at most $M+1$ positive components. □

Proof of Lemma 7. For every $i \in \mathcal{M}$ and $k = |\mathcal{S}_1^D| + 1, ..., |\mathcal{S}_M^D|$, define the binary variable $y_{i,k} = \mathbf{1}\{C_i \geq k\}$ and the marginal revenue weight $w_{i,k} = \lambda_i(R_i^*(k) - R_i^*(k-1))$. Then, for every $\boldsymbol{C} \in \mathcal{D}^M$, $F(\boldsymbol{C})$ can be represented by

$$F(\boldsymbol{C}) = \sum_{i\in\mathcal{M}} \lambda_i R_i^*(|\mathcal{S}_1^D|) + \sum_{i\in\mathcal{M}} \sum_{k=|\mathcal{S}_1^D|+1}^{|\mathcal{S}_M^D|} w_{i,k} y_{i,k}.$$

Moreover, $U_i(\boldsymbol{C})$ can also be expressed by $\boldsymbol{y}$. Specifically, for any ordered pair (m, i), we have

$$(C_m - C_i - \alpha_{m,i})^+ = \sum_{k=\alpha_{m,i}+|\mathcal{S}_1^D|+1}^{|\mathcal{S}_M^D|} \mathbf{1}\{C_m \geq k\}\mathbf{1}\{C_i < k - \alpha_{m,i}\} = \sum_{k=\alpha_{m,i}+|\mathcal{S}_1^D|+1}^{|\mathcal{S}_M^D|} y_{m,k}\left(1 - y_{i,k-\alpha_{m,i}}\right).$$

The sum is interpreted as zero when $\alpha_{m,i} + |\mathcal{S}_1^D| + 1 > |\mathcal{S}_M^D|$. Therefore, the objective function of $\Phi(\boldsymbol{\mu})$ can be represented by

$$\sum_{i\in\mathcal{M}} \lambda_i R_i^*(|\mathcal{S}_1^D|) + \sum_{i\in\mathcal{M}} \sum_{k=|\mathcal{S}_1^D|+1}^{|\mathcal{S}_M^D|} w_{i,k} y_{i,k} - \sum_{i\in\mathcal{M}} \mu_i \sum_{m\in\mathcal{M},\ m\neq i} \sum_{k=\alpha_{m,i}+|\mathcal{S}_1^D|+1}^{|\mathcal{S}_M^D|} y_{m,k}\big(1 - y_{i,k-\alpha_{m,i}}\big)$$

The consecutive relation results from the similar analysis of Constraint (5). Conversely, any binary vector satisfying these consecutive relations defines $C_i = |\mathcal{S}_1^D| + \sum_{k=|\mathcal{S}_1^D|+1}^{|\mathcal{S}_M^D|} y_{i,k}$ in $\mathcal{D}^M$, and its entries satisfy $y_{i,k} = \mathbf{1}\{C_i \geq k\}$. The same identities then preserve the objective value, establishing equivalence in both directions. □

Proof of Theorem 6. We first prove the equivalence between Problem (11) and the minimum s-t cut problem solved by Algorithm 2. Fix any $\mu \in \mathbb{R}_+^M$.

We next show that no minimum s-t cut crosses an arc of cost B. Consider the cut whose source side contains only s. Its cost is exactly W^+, because it cuts every source arc $s \to v_{i,k}$ and no other arc. Therefore, the minimum-cut value is at most W^+. Since $B = W^+ + 1$, any cut crossing an arc $v_{i,k} \to v_{i,k-1}$ of cost B has cost at least $B > W^+$ and cannot be a minimum cut.

We then establish a one-to-one correspondence between binary vectors satisfying the consecutive relations and cuts that do not cross any arc of cost B. Given such a binary vector $\boldsymbol{y}$, define the source side of the cut by

$$\mathcal{A}(\boldsymbol{y}) = \{s\} \cup \{v_{i,k} : y_{i,k} = 1\}.$$

For each prefix arc with $k = |\mathcal{S}_1^D| + 2, \ldots, |\mathcal{S}_M^D|$, if $v_{i,k} \in \mathcal{A}(\boldsymbol{y})$, then $y_{i,k} = 1$, and the consecutive relation implies $y_{i,k-1} = 1$; hence $v_{i,k-1} \in \mathcal{A}(\boldsymbol{y})$. Therefore, the cut does not cross any arc of cost B. Conversely, given a cut $(\mathcal{A}, \mathcal{V}_\mu \setminus \mathcal{A})$ that crosses no arc of cost B, define $y_{i,k} = 1$ if and only if $v_{i,k} \in \mathcal{A}$. For each such prefix arc, if $y_{i,k} = 1$, then $v_{i,k} \in \mathcal{A}$; because the arc $v_{i,k} \to v_{i,k-1}$ is not crossed, $v_{i,k-1} \in \mathcal{A}$, and therefore $y_{i,k-1} = 1$. Thus, the resulting binary vector satisfies all consecutive relations. These two constructions can be inverses of each other and maintain feasibility.

We next show that the cut cost and the maximization objective differ only by a constant based on the same correspondence as above. Recall for a binary vector $\boldsymbol{y}$ and its corresponding cut $\mathcal{A}(\boldsymbol{y})$, the source and sink arcs added in Step 1.2 contribute $W^+ - \sum_{i\in\mathcal{M}} \sum_{k=|\mathcal{S}_1^D|+1}^{|\mathcal{S}_M^D|} w_{i,k} y_{i,k}$. Moreover,

a fairness arc $v_{m,k} \to v_{i,k-\alpha_{m,i}}$ is crossed if and only if $y_{m,k}=1$ and $y_{i,k-\alpha_{m,i}}=0$. Therefore, all fairness arcs together contribute $\sum_{i,m\in\mathcal{M}} \mu_i \sum_{k=\alpha_{m,i}+|\mathcal{S}_1^D|+1}^{|\mathcal{S}_M^D|} y_{m,k}\big(1-y_{i,k-\alpha_{m,i}}\big)$. Because no arc of cost B is crossed, the total cut cost satisfies

$$c_\mu\big(A(\boldsymbol{y})\big) = W^+ - \sum_{i\in\mathcal{M}} \sum_{k=|\mathcal{S}_1^D|+1}^{|\mathcal{S}_M^D|} w_{i,k}y_{i,k} + \sum_{i\in\mathcal{M}} \mu_i \sum_{m\in\mathcal{M},\ m\neq i} \sum_{k=\alpha_{m,i}+|\mathcal{S}_1^D|+1}^{|\mathcal{S}_M^D|} y_{m,k}\big(1-y_{i,k-\alpha_{m,i}}\big).$$

Rearranging this equality gives

$$\sum_{i\in\mathcal{M}} \lambda_i R_i^*(|\mathcal{S}_1^D|) + \sum_{i\in\mathcal{M}} \sum_{k=|\mathcal{S}_1^D|+1}^{|\mathcal{S}_M^D|} w_{i,k}y_{i,k} - \sum_{i\in\mathcal{M}} \mu_i \sum_{m\in\mathcal{M},\ m\neq i} \sum_{k=\alpha_{m,i}+|\mathcal{S}_1^D|+1}^{|\mathcal{S}_M^D|} y_{m,k}\big(1-y_{i,k-\alpha_{m,i}}\big) = \sum_{i\in\mathcal{M}} \lambda_i R_i^*(|\mathcal{S}_1^D|) + W^+ - c_\mu\big(A(\boldsymbol{y})\big).$$

The left side function is the objective function of $\Phi(\mu)$. The first two terms on the right-hand side are independent of $\boldsymbol{y}$. Hence, a cut has minimum cost if and only if its corresponding binary vector maximizes the expression defining $\Phi(\mu)$.

We can now conclude the correctness of Algorithm 2. Let $(A_\mu, V_\mu \setminus A_\mu)$ be the minimum cut returned in Step 2 and define $y_{i,k}^\mu = \mathbf{1}\{v_{i,k}\in\mathcal{A}_\mu\}$. The preceding equivalence shows that $\boldsymbol{y}^\mu$ maximizes the binary formulation recalled above. Since the selected levels are consecutive, Step 3 successfully recovers the corresponding cardinality vector from the definition $y_{i,k}=\mathbf{1}_{C_i\geq k}$ as $C_i^\mu = |\mathcal{S}_1^D| + \sum_{k=|\mathcal{S}_1^D|+1}^{|\mathcal{S}_M^D|} y_{i,k}^\mu,\ i\in\mathcal{M}$. Step 3 also returns $\Phi(\boldsymbol{\mu}) = \sum_{i\in\mathcal{M}} \lambda_i R_i^*(|\mathcal{S}_1^D|) + W^+ - c_\mu(A_\mu)$ by the same cost identity.

It remains to analyze the running time. The network contains $M(|\mathcal{S}_M^D| - |\mathcal{S}_1^D|) + 2 = O(MN)$ nodes. Steps 1.2 and 1.3 add $O(MN)$ arcs, while Step 1.4 adds at most $M^2(|\mathcal{S}_M^D| - |\mathcal{S}_1^D|) = O(M^2N)$ arcs. Hence, $|V_\mu| = O(MN)$ and $|E_\mu| = O(M^2N)$. Using Dinitz's maximum-flow algorithm to compute the minimum s-t cut takes $O(|V_\mu|^2|E_\mu|)$ time (Dinitz 2006), which requires $O\big((MN)^2(M^2N)\big) = O(M^4N^3)$ time. The graph construction and the recovery of C^μ take $O(M^2N)$ and $O(MN)$ time, respectively, and are dominated by the minimum-cut computation. Therefore, Algorithm 2 solves $\Phi(\mu)$ exactly in $O(M^4N^3)$ time. □

Proof of Proposition 2. Correctness of Algorithm A.5. Algorithm A.5 applies the ellipsoid method with the separation oracle in Algorithm 2 to solve the dual problem. At iteration t, the current ellipsoid is

$$\mathcal{E}^t = \left\{\boldsymbol{y} : (\boldsymbol{y}-\boldsymbol{x}^t)^\top (\boldsymbol{Q}^t)^{-1}(\boldsymbol{y}-\boldsymbol{x}^t) \leq 1\right\}.$$

The algorithm either updates the current center x^t because its objective value cannot improve the best dual-feasible value found so far, separates the center from the dual feasible region using a violated constraint, or certifies that the center is dual feasible and updates the best feasible solution. In every case, it constructs a central halfspace containing every feasible point whose objective value is strictly smaller than the best value found so far. The next ellipsoid contains the intersection of the current ellipsoid and this halfspace.

Step 1: Definition of the initial ellipsoid. Recall that $d = M+1$, $\boldsymbol{x} = (\eta, \mu_1, \ldots, \mu_M)^\top$, and $\boldsymbol{q} = (1, \kappa_1, \ldots, \kappa_M)^\top$. For every $\boldsymbol{C}\in\mathcal{D}^M$, let $\boldsymbol{a}(\boldsymbol{C}) = (1, U_1(\boldsymbol{C}), \ldots, U_M(\boldsymbol{C}))^\top$ and $b(\boldsymbol{C}) = F(\boldsymbol{C})$. The dual feasible region is therefore

$$\mathcal{P} = \left\{\boldsymbol{x}\in\mathbb{R}^d : \boldsymbol{x}\geq\mathbf{0},\ \boldsymbol{a}(\boldsymbol{C})^\top\boldsymbol{x} \geq b(\boldsymbol{C}) \text{ for every } \boldsymbol{C}\in\mathcal{D}^M\right\},$$

and the dual objective is $\min_{\boldsymbol{x}\in\mathcal{P}} \boldsymbol{q}^\top\boldsymbol{x}$.

Let $\boldsymbol{C}^0 = (|\mathcal{S}_1^D|, \ldots, |\mathcal{S}_1^D|)$. Since $\boldsymbol{C}^0\in\mathcal{D}^M$ and $U_i(\boldsymbol{C}^0) = 0$ for every $i\in\mathcal{M}$, assigning probability one to $\boldsymbol{C}^0$ is feasible for the primal. Moreover, the dual constraint indexed by $\boldsymbol{C}^0$ implies $\eta \geq F(\boldsymbol{C}^0) \geq 0$. Thus, imposing $\eta\geq 0$ does not change the dual feasible region.

Define $\overline{F} = \sum_{i\in\mathcal{M}} \lambda_i R_i^*(|\mathcal{S}_i^D|)$. Since $F(\boldsymbol{C}) \leq \overline{F}$ for every $\boldsymbol{C}\in\mathcal{D}^M$, the point $(\overline{F}, 0, \ldots, 0)^\top$ is dual feasible. Hence, the primal and dual are both feasible and bounded, and strong duality applies.

We next justify the bounded search region. Because the cardinalities and tolerance levels are integer-valued, $U_i(\boldsymbol{C})$ is a nonnegative integer for every i and $\boldsymbol{C}$. Starting from any optimal dual solution, replace each coordinate exceeding $\overline{F}$ by $\overline{F}$. Reducing η to $\overline{F}$ preserves feasibility because $\eta = \overline{F}$ alone satisfies every cardinality constraint. If $\mu_i > \overline{F}$, reducing it to $\overline{F}$ also preserves feasibility: when $U_i(\boldsymbol{C}) = 0$, the constraint indexed by $\boldsymbol{C}$ is unchanged, whereas when $U_i(\boldsymbol{C}) > 0$, integrality gives $U_i(\boldsymbol{C}) \geq 1$ and we have $\overline{F}U_i(\boldsymbol{C}) \geq \overline{F} \geq F(\boldsymbol{C})$. Since all coefficients in the dual objective are nonnegative, this operation does not increase the objective. Consequently, there exists an optimal dual solution in $[0, B_{\mathrm{dual}}]^d$, where $B_{\mathrm{dual}} = \max\{1, \overline{F}\}$.

Algorithm A.5 initializes

$$\boldsymbol{x}^0 = \frac{B_{\mathrm{dual}}}{2}\mathbf{1}_d, \qquad \boldsymbol{Q}^0 = \frac{dB_{\mathrm{dual}}^2}{4} I_d.$$

For every $\boldsymbol{y} \in [0, B_{\mathrm{dual}}]^d$,

$$(\boldsymbol{y} - \boldsymbol{x}^0)^\top (\boldsymbol{Q}^0)^{-1} (\boldsymbol{y} - \boldsymbol{x}^0) = \frac{4\|\boldsymbol{y} - \boldsymbol{x}^0\|_2^2}{dB_{\mathrm{dual}}^2} \leq 1.$$

Thus, the initial ellipsoid $\mathcal{E}^0 = \{\boldsymbol{y} : (\boldsymbol{y} - \boldsymbol{x}^0)^\top (\boldsymbol{Q}^0)^{-1} (\boldsymbol{y} - \boldsymbol{x}^0) \leq 1\}$ contains at least one optimal dual solution.

Step 2: The separation oracle. For any $\boldsymbol{\mu} \geq \mathbf{0}$, the exponentially many cardinality constraints are equivalent to

$$\eta \geq \max_{\boldsymbol{C} \in \mathcal{D}^M} \left\{ F(\boldsymbol{C}) - \sum_{i \in \mathcal{M}} \mu_i U_i(\boldsymbol{C}) \right\} = \Phi(\boldsymbol{\mu}).$$

Algorithm 2 returns an exact maximizer $\boldsymbol{C}^t$ and the exact value $\Phi(\boldsymbol{\mu}^t) = F(\boldsymbol{C}^t) - \sum_{i \in \mathcal{M}} \mu_i^t U_i(\boldsymbol{C}^t)$ in polynomial time. If $\Phi(\boldsymbol{\mu}^t) > \eta^t$, then $\boldsymbol{a}(\boldsymbol{C}^t)^\top \boldsymbol{x}^t = \eta^t + \sum_{i \in \mathcal{M}} \mu_i^t U_i(\boldsymbol{C}^t) < F(\boldsymbol{C}^t) = b(\boldsymbol{C}^t)$, so the constraint indexed by $\boldsymbol{C}^t$ is violated. If $\Phi(\boldsymbol{\mu}^t) \leq \eta^t$, then $F(\boldsymbol{C}) - \sum_{i \in \mathcal{M}} \mu_i^t U_i(\boldsymbol{C}) \leq \Phi(\boldsymbol{\mu}^t) \leq \eta^t$ for every $\boldsymbol{C} \in \mathcal{D}^M$, and hence $\boldsymbol{a}(\boldsymbol{C})^\top \boldsymbol{x}^t \geq b(\boldsymbol{C})$ for every $\boldsymbol{C}$. Because the oracle is invoked only after all nonnegativity conditions have been verified, $\boldsymbol{x}^t$ is dual feasible. Therefore, Algorithm 2 is an exact separation oracle, and each returned maximizer is added to $\mathcal{C}_v$.

Step 3: Interpretation of each branch of the while loop. Let $\boldsymbol{x}^{\mathrm{best}}$ be the best dual solution found before iteration t, and let $\mathrm{OBJ} = \boldsymbol{q}^\top \boldsymbol{x}^{\mathrm{best}}$. The dual solutions that can strictly improve this value form the set

$$\mathcal{K}^t = \left\{ \boldsymbol{y} \in \mathcal{P} : \boldsymbol{q}^\top \boldsymbol{y} < \mathrm{OBJ} \right\}.$$

Each branch chooses a nonzero direction $\boldsymbol{g}^t$ such that $(\boldsymbol{g}^t)^\top (\boldsymbol{y} - \boldsymbol{x}^t) \geq 0$ for every $\boldsymbol{y} \in \mathcal{K}^t$. Thus, all potentially better dual solutions lie in the retained halfspace $\mathcal{H}^t = \{\boldsymbol{y} : (\boldsymbol{g}^t)^\top (\boldsymbol{y} - \boldsymbol{x}^t) \geq 0\}$.

Case 1: The center cannot improve the current best value. Suppose $\boldsymbol{q}^\top \boldsymbol{x}^t \geq \mathrm{OBJ}$. Because every $\boldsymbol{y} \in \mathcal{K}^t$ satisfies $\boldsymbol{q}^\top \boldsymbol{y} < \mathrm{OBJ} \leq \boldsymbol{q}^\top \boldsymbol{x}^t$, we have $(-\boldsymbol{q})^\top (\boldsymbol{y} - \boldsymbol{x}^t) = \boldsymbol{q}^\top \boldsymbol{x}^t - \boldsymbol{q}^\top \boldsymbol{y} > 0$. Hence we set $\boldsymbol{g}^t = -\boldsymbol{q}$.

Case 2: A nonnegativity constraint is violated. Suppose $\boldsymbol{q}^\top \boldsymbol{x}^t < \mathrm{OBJ}$ and the coordinate x_j^t is negative, where $j = 0$ corresponds to η and $j = i$ corresponds to μ_i. Every feasible solution satisfies $y_j \geq 0 > x_j^t$. Thus, with $\boldsymbol{g}^t = \boldsymbol{e}_j$, we have $(\boldsymbol{g}^t)^\top (\boldsymbol{y} - \boldsymbol{x}^t) = y_j - x_j^t > 0$ for every $\boldsymbol{y} \in \mathcal{P}$. Accordingly, the algorithm uses $\boldsymbol{g}^t = \boldsymbol{e}_0$ when $\eta^t < 0$ and $\boldsymbol{g}^t = \boldsymbol{e}_{i^t}$ when $\mu_{i^t}^t < 0$.

Case 3: The oracle finds a violated cardinality constraint. Suppose $\boldsymbol{x}^t \geq \mathbf{0}$ and $\Phi(\boldsymbol{\mu}^t) > \eta^t$. Then $\boldsymbol{a}(\boldsymbol{C}^t)^\top \boldsymbol{x}^t < b(\boldsymbol{C}^t)$, whereas every dual-feasible solution $\boldsymbol{y}$ satisfies $\boldsymbol{a}(\boldsymbol{C}^t)^\top \boldsymbol{y} \geq b(\boldsymbol{C}^t)$. Hence $\boldsymbol{a}(\boldsymbol{C}^t)^\top (\boldsymbol{y} - \boldsymbol{x}^t) > 0$ for every $\boldsymbol{y} \in \mathcal{P}$, so $\boldsymbol{g}^t = \boldsymbol{a}(\boldsymbol{C}^t)$ is a valid cut direction. The algorithm records $\boldsymbol{C}^t$ in $\mathcal{C}_v$ because this constraint will be used to recover an optimal primal solution.

Case 4: No cardinality constraint is violated. Suppose $\Phi(\boldsymbol{\mu}^t) \le \eta^t$. Then $\boldsymbol{x}^t$ is dual feasible. Since this branch is reached only after Case 1 has failed, $\boldsymbol{q}^\top \boldsymbol{x}^t < \mathrm{OBJ}$, so the algorithm updates $\boldsymbol{x}^{\mathrm{best}} \leftarrow \boldsymbol{x}^t$ and $\mathrm{OBJ} \leftarrow \boldsymbol{q}^\top \boldsymbol{x}^t$. The algorithm therefore sets $\boldsymbol{g}^t = -\boldsymbol{q}$ to search for a better feasible solution.

In all four cases, $\mathcal{H}^t$ contains every dual feasible solution whose objective value is strictly smaller than the best value after any update in iteration t. If no such point exists, the current best feasible solution is optimal; otherwise, the algorithm continues shrinking the ellipsoid.

Step 4: Ellipsoid update. The halfspace $\mathcal{H}^t$ passes through the center $\boldsymbol{x}^t$, so it defines a central cut of $\mathcal{E}^t$. Applying the standard central-cut update in Bertsimas and Tsitsiklis (1997) gives

$$\boldsymbol{x}^{t+1} = \boldsymbol{x}^t + \frac{1}{d+1}\frac{\boldsymbol{Q}^t\boldsymbol{g}^t}{\sqrt{(\boldsymbol{g}^t)^\top \boldsymbol{Q}^t \boldsymbol{g}^t}}, \quad \boldsymbol{Q}^{t+1} = \frac{d^2}{d^2-1}\left(\boldsymbol{Q}^t - \frac{2}{d+1}\frac{\boldsymbol{Q}^t\boldsymbol{g}^t(\boldsymbol{g}^t)^\top \boldsymbol{Q}^t}{(\boldsymbol{g}^t)^\top \boldsymbol{Q}^t \boldsymbol{g}^t}\right),$$

where $d = M+1$. The positive sign in the center update is consistent with retaining the side $(\boldsymbol{g}^t)^\top(\boldsymbol{y} - \boldsymbol{x}^t) \ge 0$. These formulas describe the central-cut update.

Running time of Algorithm A.5. The preliminary computation of all $R_i^*(C)$ and $\mathcal{S}_i^*(C)$ is polynomial-time, as established earlier. In each ellipsoid iteration, the objective and nonnegativity checks require $O(M)$ operations, and the ellipsoid update requires $O(M^2)$ operations. At most one call to compute $\Phi(\boldsymbol{\mu}^t)$ is made, which is in polynomial time. Let L be the binary encoding length of the original rational input. Every $R_i^*(C)$ is the ratio of sums and products of at most N input coefficients, so its encoding length is polynomial in N and L. Also, $0 \le U_i(\boldsymbol{C}) \le (M-1)N$, and $U_i(\boldsymbol{C})$ is integer-valued. Thus a common polynomial bound on the encoding lengths of $\boldsymbol{a}(\boldsymbol{C})$, $b(\boldsymbol{C})$, $\boldsymbol{q}$, and B_{dual} is available without enumerating $\mathcal{D}^M$. The feasible region is therefore a well-described rational polyhedron. It has a finite optimum attained in the bounded region established in Step 1, and Step 2 supplies strong separation in polynomial time. The result in Grötschel et al. (1993) therefore applies: Including rational rounding, any auxiliary feasibility calls, and exact termination, it returns an exact optimal dual solution in time polynomial in M, N, L. Algorithm A.5 uses this complete procedure.

Size of $\mathcal{C}_v$. The complete procedure just established makes polynomially many oracle calls, including those used for exact termination and solution recovery. Since $\mathcal{C}_v$ initially contains only $\boldsymbol{C}^0$ and each call records at most one maximizing cardinality vector, its final size is polynomial in M, N, L.

□

Proof of Theorem 7. We first establish that restricting the primal to $\mathcal{C}_v$ preserves the optimal value, and then prove optimality and polynomial-time recovery of the stated assortment policy.

(i) Equality of the full and restricted optimal values. Let Z^* denote the optimal value of the full dual. Consider the restricted dual

$$Z_v = \min_{\eta\in\mathbb{R},\,\boldsymbol{\mu}\ge\boldsymbol{0}} \quad \eta + \sum_{i\in\mathcal{M}} \kappa_i\mu_i \quad \text{s.t.} \quad \eta + \sum_{i\in\mathcal{M}} \mu_i U_i(\boldsymbol{C}) \ge F(\boldsymbol{C}), \quad \boldsymbol{C}\in\mathcal{C}_v.$$

Because $\boldsymbol{C}^0 \in \mathcal{C}_v$, this restricted dual also implies $\eta \ge F(\boldsymbol{C}^0) \ge 0$ and therefore has the same variable region used by Algorithm A.5. Since it retains only a subset of the full constraints, we have $Z_v \le Z^*$.

Observe that every feasibility cut generated during the iterations is either a nonnegativity cut or a cardinality constraint indexed by a vector in $\mathcal{C}_v$, while every objective cut depends only on $\boldsymbol{q}$ and the best feasible value found so far. Use the same initialization, B_{dual} as well as other computing details for both dual problems.

Consider any cardinality-oracle call in the complete full-dual execution and denote its multiplier vector by $\widehat{\boldsymbol{\mu}}$. Let $\widehat{\boldsymbol{C}}$ be the returned maximizer. By construction, $\widehat{\boldsymbol{C}} \in \mathcal{C}_v \subseteq \mathcal{D}^M$, so

$$\max_{\boldsymbol{C}\in\mathcal{C}_v}\{F(\boldsymbol{C}) - \widehat{\boldsymbol{\mu}}^\top \boldsymbol{U}(\boldsymbol{C})\} \le \max_{\boldsymbol{C}\in\mathcal{D}^M}\{F(\boldsymbol{C}) - \widehat{\boldsymbol{\mu}}^\top \boldsymbol{U}(\boldsymbol{C})\} = F(\widehat{\boldsymbol{C}}) - \widehat{\boldsymbol{\mu}}^\top \boldsymbol{U}(\widehat{\boldsymbol{C}}) \le \max_{\boldsymbol{C}\in\mathcal{C}_v}\{F(\boldsymbol{C}) - \widehat{\boldsymbol{\mu}}^\top \boldsymbol{U}(\boldsymbol{C})\},$$

where $\boldsymbol{U}(\boldsymbol{C}) = (U_i(\boldsymbol{C}))_{i\in\mathcal{M}}$. Hence the two oracle values coincide. The fixed total order selects the same maximizer because the full-dual choice is retained. It follows inductively that every oracle

query and answer, nonnegativity and objective check, incumbent update, and rational ellipsoid update is identical in the two executions. The exact stopping and recovery operations therefore produce the same exact output. Since the procedure solves each dual exactly, $Z_v = Z^*$.

Strong duality for the restricted pair yields

$$\mathrm{OPT}(\text{Problem (14)}) = Z_v = Z^* = \mathrm{OPT}(\text{Problem (9)}).$$

Moreover, every optimal solution $\boldsymbol{p}^* = (p^*_{\boldsymbol{C}})_{\boldsymbol{C}\in\mathcal{C}_v}$ to Problem (14) can be extended to a solution to Problem (9) by retaining $p^*_{\boldsymbol{C}}$ for every $\boldsymbol{C} \in \mathcal{C}_v$ and assigning zero probability to every $\boldsymbol{C} \in \mathcal{D}^M \setminus \mathcal{C}_v$. Clearly, it is feasible and attains the optimal objective value. Therefore, it is optimal for Problem (9).

(ii) Optimality of the recovered assortment policy. We next recover an optimal assortment policy. By Proposition 1 and the fixed-cardinality reformulation in Section 5.2, Problems (RFA-M) and (9) have the same optimal value. Part (i) above further shows that extending $\boldsymbol{p}^*$ by zero outside $\mathcal{C}_v$ yields an optimal solution to Problem (9). For every $\boldsymbol{C} \in \mathcal{C}_v$, the associated assortment tuple $\big(\mathcal{S}^*_i(C_i)\big)_{i\in\mathcal{M}}$ satisfies $|\mathcal{S}^*_i(C_i)| = C_i$ and $R_i(\mathcal{S}^*_i(C_i)) = R^*_i(C_i)$. Thus, assigning probability $p^*_{\boldsymbol{C}}$ to this tuple preserves both its revenue $F(\boldsymbol{C})$ and every unfairness term $U_i(\boldsymbol{C})$. Consequently, G^F is feasible for Problem (RFA-M) and attains its optimal value $\sum_{\boldsymbol{C}\in\mathcal{C}_v} p^*_{\boldsymbol{C}} F(\boldsymbol{C})$.

(iii) Running time and solution recovery. We next establish the running time. Lemma 3 guarantees that all required fixed-cardinality revenues and assortments can be computed in polynomial time. Theorem 6, based on Lemma 7, provides an exact polynomial-time oracle for $\Phi(\boldsymbol{\mu})$. Hence, Proposition 2 guarantees polynomial-time solution of the dual problem. At most one cardinality vector is recorded in each oracle call of the complete exact procedure, so $|\mathcal{C}_v|$ is polynomially bounded in M, N, L. Problem (14) therefore has polynomially many variables, M fairness constraints, and one normalization constraint. Its coefficients can be constructed in polynomial time and have polynomial binary encoding length, so an optimal solution $\boldsymbol{p}^*$ can also be computed exactly in polynomial time.

Finally, choosing $\boldsymbol{p}^*$ to be an optimal basic solution yields at most $M+1$ positive probabilities by the same argument used in the proof of Proposition 1. The corresponding assortment tuples are recovered from the precomputed table, while all other probabilities are implicitly zero. Therefore, an optimal policy G^F can be obtained exactly in polynomial time. □

Appendix C.2. Proofs of Section 6

Proof of Lemma 8: We only need to prove that for any $C_1 \le C_2$, we have $\frac{R^*_i(C_1)}{C_1} \ge \frac{R^*_i(C_2)}{C_2}$. Recall that

$$R^*_i(C) = \max_{S:|S|=C} \frac{\sum_{j\in S} r_j v_{ij}}{1+\sum_{j\in S} v_{ij}} = \max_R R \text{ s.t. } \max_{S:|S|=C} \sum_{j\in S} h_{ij}(R) \ge R,$$

where $h_{ij}(R) = (r_j - R)v_{ij}$ and $\max_{S:|S|=C}\sum_{j\in S} h_{ij}(R)$ is equivalent to the summation of the top C value of $h_{ij}(R)$, $j \in \mathcal{N}$ for each R. Denote the Cth largest $h_{ij}(R)$ under a fixed R as $h^{(C)}_j(R)$.

For any $R \ge 0$, we have

$$\begin{aligned}
\max_{S:|S|=C_2}\sum_{j\in S} h_{ij}(R) &\le \max_{S:|S|=C_1}\sum_{j\in S} h_{ij}(R) + (C_2-C_1)h^{(C_1)}_j(R)\\
&\le \max_{S:|S|=C_1}\sum_{j\in S} h_{ij}(R) + \tfrac{(C_2-C_1)}{C_1}C_1 h^{(C_1)}_j(R)\\
&\le \max_{S:|S|=C_1}\sum_{j\in S} h_{ij}(R) + \tfrac{(C_2-C_1)}{C_1}\max_{S:|S|=C_1}\sum_j h^{(C_1)}_j(R)\\
&= \tfrac{C_2}{C_1}\max_{S:|S|=C_1}\sum_j h^{(C_1)}_j(R),
\end{aligned}$$

where the first and third inequality are because $h^{(C)}_j(R)$ decreases in C and $C_1 \le C_2$. It leads to the following inequality

$$\begin{aligned}
R^*_i(C_2) &= \max_R R \text{ s.t. } \max_{S:|S|=C_2}\sum_{j\in S}(r_j - R)v_{ij} \ge R\\
&\le \max_R R \text{ s.t. } \tfrac{C_2}{C_1}\max_{S:|S|=C_1}\sum_{j\in S} h_{ij}(R) \ge R.
\end{aligned}$$

Set the variable $x = \frac{C_1}{C_2}R$. Then we have

$$\begin{aligned} R_i^*(C_2) &\le \tfrac{C_2}{C_1}\max_x x \text{ s.t. } \tfrac{C_2}{C_1}\max_{S:|S|=C_1}\textstyle\sum_{j\in S} h_{ij}(\tfrac{C_2}{C_1}x) \ge \tfrac{C_2}{C_1}x \\ &= \tfrac{C_2}{C_1}\max_x x \text{ s.t. } \max_{S:|S|=C_1}\textstyle\sum_{j\in S} h_{ij}(\tfrac{C_2}{C_1}x) - x \ge 0 \\ &\le \tfrac{C_2}{C_1}\max_x x \text{ s.t. } \max_{S:|S|=C_1}\textstyle\sum_{j\in S} h_{ij}(x) - x \ge 0 \\ &= \tfrac{C_2}{C_1}R_i^*(C_1), \end{aligned}$$

where the last inequality is because $h_{ij}(x)$ decreases in x for all j. This completes the proof. □

Proof of Proposition 3. (i) The first bound: Select $h \in \arg\max_i \lambda_i R_i^*(|\mathcal{S}_i^D|)$. Set $\mathcal{S}_i^F = \mathcal{S}_i^*(|\mathcal{S}_h^D|)$ for all $i \in \mathcal{M}$. Then the assortment is feasible and we have $\Pi_S^F/\Pi^D \ge 1/M$.
(ii) The second bound: Recall that $D_i^U = \min_{j\in\mathcal{M}}\{D_j + d_{i,j}\}$, where $D_i = |\mathcal{S}_i^D|$. Since $d_{i,j} \ge 0$, $d_{i,i} = 0$, and $D_j \ge D_1$, we have $D_1 \le D_i^U \le D_i$ for every $i \in \mathcal{M}$.

We first show that the assortment with the cardinality vector $\boldsymbol{C} = \boldsymbol{D}^U$ is feasible for the original fairness constraints. Fix any distinct $i, j \in \mathcal{M}$. The triangle inequality for the shortest-path distances gives

$$D_i^U = \min_{m\in\mathcal{M}}\{D_m + d_{i,m}\} \le \min_{m\in\mathcal{M}}\{D_m + d_{j,m} + d_{i,j}\} = D_j^U + d_{i,j} \le D_j^U + \alpha_{i,j}.$$

which implies $D_i^{\mathrm{U}} - D_j^{\mathrm{U}} \le \alpha'_{i,j}$. Therefore, such assortment decision with cardinality $\boldsymbol{D}^{\mathrm{U}}$ is feasible.

Therefore, the optimal revenue satisfies $\Pi_S^F \ge \sum_{i\in\mathcal{M}} \lambda_i R_i^*(D_i^{\mathrm{U}})$. Note that Lemma 8 implies $R_i^*(D_i^{\mathrm{U}})/D_i^{\mathrm{U}} \ge R_i^*(|\mathcal{S}_i^D|)/|\mathcal{S}_i^D|$. Therefore,

$$\Pi_{\mathcal{S}}^F \ge \sum_{i\in\mathcal{M}} \lambda_i R_i^*(D_i)\frac{D_i^{\mathrm{U}}}{|\mathcal{S}_i^D|}.$$

Dividing by Π^D gives

$$\frac{\Pi_{\mathcal{S}}^F}{\Pi^D} \ge \sum_{i\in\mathcal{M}} \frac{\lambda_i R_i^*(D_i)}{\Pi^D}\frac{D_i^{\mathrm{U}}}{|\mathcal{S}_i^D|} \ge \min_{i\in\mathcal{M}}\frac{D_i^{\mathrm{U}}}{|\mathcal{S}_i^D|},$$

where the last inequality is because $\Pi^D = \sum_{i\in\mathcal{M}} \lambda_i R_i^*(D_i)$.
(iii) The last bound: Let $\omega_i = \lambda_i R_i^*(|\mathcal{S}_i^D|)/\Pi^D, \quad i \in \mathcal{M}$. Thus, ω_i is type i's share of the fully customized revenue, and $\sum_{i\in\mathcal{M}}\omega_i = 1$. Clearly, $\omega_i \le \Pi_{\mathcal{S}}^F/\Pi^D$ for every $i \in \mathcal{M}$. Recall from part (ii) that $\boldsymbol{D}^U = (D_i^U)_{i\in\mathcal{M}}$ is a feasible cardinality vector and $R_i^*(D_i^{\mathrm{U}})/R_i^*(|\mathcal{S}_i^D|) \ge D_i^{\mathrm{U}}/|\mathcal{S}_i^D|$. Consequently,

$$\frac{\Pi_{\mathcal{S}}^F}{\Pi^D} \ge \frac{\sum_{i\in\mathcal{M}}\lambda_i R_i^*(D_i^U)}{\Pi^D} \ge \sum_{i\in\mathcal{M}}\omega_i\frac{D_i^U}{|\mathcal{S}_i^D|} = 1 - \sum_{i\in\mathcal{M}}\omega_i\left(1 - \frac{D_i^U}{|\mathcal{S}_i^D|}\right) \ge 1 - \frac{\Pi_{\mathcal{S}}^F}{\Pi^D}\sum_{i\in\mathcal{M}}\left(1 - \frac{D_i^U}{|\mathcal{S}_i^D|}\right).$$

Rearranging the last inequality yields

$$\frac{\Pi_{\mathcal{S}}^F}{\Pi^D} \ge \frac{1}{1 + \sum_{i\in\mathcal{M}}\left(1 - \frac{D_i^U}{|\mathcal{S}_i^D|}\right)}.$$

In addition, if the fully customized assortment satisfies all fairness constraints, then $D_i \le D_j + d_{i,j}$ for every $i, j \in \mathcal{M}$. Since $d_{i,i} = 0$, this gives $D_i^U = D_i = |\mathcal{S}_i^D|$ for every $i \in \mathcal{M}$. Hence the second term of our lower bound will become 1, which leads to $L_S = 1$. □

Proof of Proposition 4. Let $(\mathcal{S}_i^F)_{i\in\mathcal{M}}$ be an optimal deterministic assortment strategy for Problem (DFA-M) with tolerance level $\boldsymbol{\alpha}$. Consider the randomized policy that offers $(\mathcal{S}_i^D)_{i\in\mathcal{M}}$ with probability p^0 and $(\mathcal{S}_i^F)_{i\in\mathcal{M}}$ with probability $1-p^0$.

Because $(\mathcal{S}_i^F)_{i\in\mathcal{M}}$ has zero excess unfairness, the expected unfairness perceived by type i under this policy is

$$p^0 \sum_{m\in\mathcal{M}} \left(|\mathcal{S}_m^D| - |\mathcal{S}_i^D| - \alpha_{m,i}\right)^+ \le \kappa_i,$$

where the inequality follows from the definition of p^0. Hence the policy is feasible for Problem (RFA-M). Its expected revenue is $p^0\Pi^D + (1-p^0)\Pi_{\mathcal{S}}^F$. Therefore, the optimality of G^F and Proposition 3 imply

$$\frac{\Pi_G^F}{\Pi^D} \ge p^0 + (1-p^0)\frac{\Pi_{\mathcal{S}}^F}{\Pi^D} \ge p^0 + (1-p^0)L_S.$$

If the fully customized policy satisfies all randomized fairness constraints, then $\sum_{m\in\mathcal{M}}\left(|\mathcal{S}_m^D| - |\mathcal{S}_i^D| - \alpha_{m,i}\right)^+ \le \kappa_i$ for every $i\in\mathcal{M}$, so $p^0=1$ and the lower bound will become 1. □

Proof of Proposition 5. (i) Lower bound: The feasible policy constructed in the proof of Proposition 4 gives $\Pi_G^F \ge p^0\Pi^D + (1-p^0)\Pi_{\mathcal{S}}^F$. Dividing both sides by $\Pi_{\mathcal{S}}^F$ yields

$$\frac{\Pi_G^F}{\Pi_{\mathcal{S}}^F} \ge 1 + p^0\left(\frac{\Pi^D}{\Pi_{\mathcal{S}}^F} - 1\right).$$

(ii) Upper bound: First, no randomized policy can earn more than the fully customized policy, so $\Pi_G^F \le \Pi^D$, and hence $\Pi_G^F/\Pi_{\mathcal{S}}^F \le \Pi^D/\Pi_{\mathcal{S}}^F$. Here, we use the shortest-path distances $d_{i,j}$ from Appendix A.2 to prove the result.

Suppose we select an optimal randomized policy such that every cardinality vector $\boldsymbol{C}$ in its support belongs to $\mathcal{D}^M$. Following the definition $D_i^U = \min_{j\in\mathcal{M}}\{|\mathcal{S}_j^D| + d_{i,j}\}$, for every such $\boldsymbol{C}$, define

$$C_i^U := \min_{j\in\mathcal{M}}\{C_j + d_{i,j}\} \in [|\mathcal{S}_1^D|, C_i], \qquad i\in\mathcal{M}.$$

Similar to the proof in Proposition 3, $\boldsymbol{C}^U = (C_i^U)_{i\in\mathcal{M}}$ is feasible for deterministic fairness problem, that is $C_i^U - C_j^U \le d_{i,j}$ for every $i,j\in\mathcal{M}$.

We next bound the reduction from C_i to C_i^U. If $C_i = C_i^U$, the required bound is immediate. Otherwise, suppose $C_i^U = C_{j^\star} + d_{i,j^\star}$, and let $i = i_0, i_1, \ldots, i_L = j^\star$ be a simple shortest path attaining $d_{i,j^\star}$. Since $d_{i,j^\star} < C_i - C_{j^\star} \le N$, every arc on this path has length equal to its original tolerance. Then

$$C_i - C_i^U = C_i - C_{j^\star} - d_{i,j^\star} = \sum_{\ell=0}^{L-1}\left(C_{i_\ell} - C_{i_{\ell+1}} - \alpha_{i_\ell,i_{\ell+1}}\right) \le \sum_{\ell=0}^{L-1}\left(C_{i_\ell} - C_{i_{\ell+1}} - \alpha_{i_\ell,i_{\ell+1}}\right)^+ \le \sum_{i\in\mathcal{M}} U_i(\boldsymbol{C}).$$

Since $C_i^U \le C_i$, Lemma 8 implies

$$R_i^*(C_i) \le \frac{C_i}{C_i^U}R_i^*(C_i^U) = \left(1 + \frac{C_i - C_i^U}{C_i^U}\right)R_i^*(C_i^U) \le \left(1 + \frac{\sum_{q\in\mathcal{M}} U_q(C)}{|\mathcal{S}_1^D|}\right)R_i^*(C_i^U).$$

After multiplying by λ_i and summing over i, we obtain

$$\sum_i \lambda_i R_i^*(C_i) \le \left(1 + \frac{\sum_{q\in\mathcal{M}} U_q(C)}{|\mathcal{S}_1^D|}\right)\sum_i \lambda_i R_i^*(C_i^U) \le \left(1 + \frac{\sum_{q\in\mathcal{M}} U_q(C)}{|\mathcal{S}_1^D|}\right)\Pi_{\mathcal{S}}^F,$$

where the last inequality is due to the feasibility of $\boldsymbol{C}^U$ to the deterministic problem. Taking expectations under G^F and applying the randomized fairness constraints yields

$$\Pi_G^F = \mathbb{E}_{G^F}[\sum_i \lambda_i R_i^*(C_i)] \le \left(1 + \frac{\sum_{i\in\mathcal{M}} \mathbb{E}_{G^F}[U_i(C)]}{|\mathcal{S}_1^D|}\right) \Pi_{\mathcal{S}}^F \le \left(1 + \frac{\sum_{q\in\mathcal{M}} \kappa_q}{|\mathcal{S}_1^D|}\right) \Pi_{\mathcal{S}}^F.$$

Dividing by Π_S^F proves the upper bound. □

Proof of Proposition 6. By Proposition 1, choose an optimal policy G^F that offers fixed-cardinality optimal assortments $\mathcal{S}_i^*(C_i)$. Fix a customer type i such that $C_i \ge |\mathcal{S}_i^D|$ in the support of G^F. We first show that $W_i(\mathcal{S}_i^*(C_i)) \ge W_i(\mathcal{S}_i^D)$. If $\mathcal{S}_i^D \subseteq \mathcal{S}_i^*(C_i)$, then $W_i(\mathcal{S}_i^*(C_i)) \ge W_i(\mathcal{S}_i^D)$ automatically holds. Otherwise, assume $\mathcal{S}_i^D \backslash \mathcal{S}_i^*(C_i) \ne \emptyset$. If $R_i(\mathcal{S}_i^*(C_i)) = R_i(\mathcal{S}_i^D)$, then $\mathcal{S}_i^*(C_i)$ is also fully customized optimal and must contain every product with $r_j > R_i(\mathcal{S}_i^D)$: adding any missing such product would strictly increase revenue. By our smallest-cardinality convention, $\mathcal{S}_i^D = \{j : r_j > R_i(\mathcal{S}_i^D)\}$, so equality would imply $\mathcal{S}_i^D \subseteq \mathcal{S}_i^*(C_i)$, a contradiction. Thus, in the case under consideration, $R_i(\mathcal{S}_i^*(C_i)) < R_i(\mathcal{S}_i^D)$. Recall that $\mathcal{S}_i^D = \arg\max_{S:|S|=|\mathcal{S}_i^D|} \sum_{j\in S} h_{ij}(R_i(\mathcal{S}_i^D))$, $\mathcal{S}_i^*(C_i) = \arg\max_{S:|S|=|\mathcal{S}_i^*(C_i)|} \sum_{j\in S} h_{ij}(R_i(\mathcal{S}_i^*(C_i)))$ and $R_i(\mathcal{S}_i^*(C_i)) < R_i(\mathcal{S}_i^D)$. Therefore, for any $j_1 \in \mathcal{S}_i^*(C_i)\backslash\mathcal{S}_i^D$ and $j_2 \in \mathcal{S}_i^D\backslash\mathcal{S}_i^*(C_i)$, we have $h_{ij_1}(R_i(\mathcal{S}_i^*(C_i))) \ge h_{ij_2}(R_i(\mathcal{S}_i^*(C_i)))$ and $h_{ij_1}(R_i(\mathcal{S}_i^D)) \le h_{ij_2}(R_i(\mathcal{S}_i^D))$. Since $h_{ij}(R) = v_{ij}(r_j - R)$ is a linear function in terms of R and $R_i(\mathcal{S}_i^*(C_i)) < R_i(\mathcal{S}_i^D)$, then we have $v_{ij_1} \ge v_{ij_2}$ for any $j_1 \in \mathcal{S}_i^*(C_i)\backslash\mathcal{S}_i^D$ and $j_2 \in \mathcal{S}_i^D\backslash\mathcal{S}_i^*(C_i)$. In addition to the fact $|\mathcal{S}_i^*(C_i)| \ge |\mathcal{S}_i^D|$, it follows that $\sum_{j\in\mathcal{S}_i^*(C_i)} v_{ij} \ge \sum_{j\in\mathcal{S}_i^D} v_{ij}$ and $W_i(\mathcal{S}_i^*(C_i)) \ge W_i(\mathcal{S}_i^D)$.

Since this inequality holds for every $\boldsymbol{C}$ in the support of G^F, taking expectation yields $\mathbb{E}_{G^F}[W_i(\mathcal{S}_i)] \ge W_i(\mathcal{S}_i^D)$.

Deterministic special case. Finally, supposing that $\kappa_i = 0$ for every $i \in \mathcal{M}$, the randomized problem reduces to the deterministic fairness problem. Similar to the above proof, for any type i satisfying $|\mathcal{S}_i^F| \ge |\mathcal{S}_i^D|$, we have $W_i(\mathcal{S}_i^F) \ge W_i(\mathcal{S}_i^D)$. □

Appendix C.3. Proof of Appendix A

Proof of Lemma A.1: We first show that Step 2 preserves the feasible region and produces tight bounds. Capping these tolerances at N does not change $\mathcal{F}_0$. Suppose we have a feasible solution under d_{i_1,i_2} to be $\mathcal{S}_i'$, $i \in \mathcal{M}$. Based on the definition, we have $|\mathcal{S}_{i_1}'| - |\mathcal{S}_{i_2}'| \le d_{i_1,i_2} \le \alpha_{i_1,i_2}$ for any $i_1, i_2 \in \mathcal{M}$. So $\mathcal{S}_i'$ is also feasible for Problem (DFA-M). On the other hand, suppose we have a feasible solution under α_{i_1,i_2} to be $\mathcal{S}_i$, $i \in \mathcal{M}$. Assume d_{i_1,i_2} is obtained by the shortest path $i_{k_1} \to i_{k_2} \to \cdots \to i_{k_K}$, where $i_{k_1} = i_1$, $i_{k_K} = i_2$ and $d_{i_1,i_2} = \sum_{l\le K-1} \min\{N, \alpha_{i_{k_l},i_{k_{l+1}}}\}$. Then for any $i_1, i_2 \in \mathcal{M}$, we have

$$|\mathcal{S}_{i_1}| - |\mathcal{S}_{i_2}| = \sum_{l\le K-1} \left(|\mathcal{S}_{i_{k_l}}| - |\mathcal{S}_{i_{k_{l+1}}}|\right) \le \sum_{l\le K-1} \min\{N, \alpha_{i_{k_l},i_{k_{l+1}}}\} = d_{i_1,i_2}.$$

Thus $\mathcal{S}_i$, $i \in \mathcal{M}$ is also feasible for the problem under d_{i_1,i_2} and the two problems are equivalent. Moreover, fix $j \in \mathcal{M}$ and set $C_k = d_{k,j}$ for every $k \in \mathcal{M}$. These cardinalities are integers in $[0, N]$, and the triangle inequality gives $C_k - C_\ell = d_{k,j} - d_{\ell,j} \le d_{k,\ell} \le \alpha_{k,\ell}$ for every distinct k, ℓ. Thus, this constructed vector $\boldsymbol{C}$ belongs to $\mathcal{F}_0$ and attains $C_i - C_j = d_{i,j}$ for every i because $d_{j,j} = 0$. Consequently,

$$d_{i,j} = \max_{\boldsymbol{C}\in\mathcal{F}_0} (C_i - C_j), \qquad i, j \in \mathcal{M}. \tag{C.10}$$

We next show that the constraints retained in Step 3 preserve $\mathcal{F}_0$. We prove that every vector satisfying the retained constraints belongs to $\mathcal{F}_0$, and conversely. Let E denote the ordered class pairs retained in Step 3.2. The class distance $d_{Q,R}$ is independent of the chosen representatives by the triangle inequality and the zero distances in both directions within each class. Take any vector

$\boldsymbol{C} \in \{0, 1, \ldots, N\}^M$ satisfying the retained constraints. The cycles in Step 3.1 force all cardinalities within each class to be equal. For distinct classes Q, R with $d_{Q,R} < N$, consider the complete directed graph on the classes, with arc lengths $d_{U,V}$, and choose a shortest Q-R path having the largest number of arcs. Its total length is $d_{Q,R}$, so every arc (U, V) on this path satisfies $0 \le d_{U,V} < N$ and is considered in Step 3.2. If some arc (U, V) were not retained, the condition in Step 3.2 and the triangle inequality would imply $d_{U,V} = d_{U,T} + d_{T,V}$ for a third class T. Replacing this arc by $U \to T \to V$ would produce a shortest walk with more arcs. This walk cannot contain a positive-length cycle, which would contradict shortestness, or a zero-length cycle, whose classes would belong to the same zero-distance equivalence class. It would therefore be a path, contradicting our choice. Thus, the retained arcs reproduce every distance below N. Summing their constraints along such a path gives $C_i - C_j \le d_{Q,R} = d_{i,j}$ for every $i \in Q$ and $j \in R$. Bounds with $d_{Q,R} = N$ follow directly from $0 \le C_i \le N$. Together with equality within each class and $d_{i,j} \le \alpha_{i,j}$, these bounds imply $\boldsymbol{C} \in \mathcal{F}_0$. Conversely, every retained constraint is valid for $\mathcal{F}_0$. Hence, the reduced and original feasible regions coincide.

We now relate any equivalent representation to the same distances to prepare the minimum-cardinality argument. Consider any feasible representation $(\mathcal{A}, \boldsymbol{x}_\mathcal{A})$ of the reduction problem. Construct a directed graph with an arc of length $x_{i,j}$ for every $(i, j) \in \mathcal{A}$, and add auxiliary arcs of length N between all distinct nodes. These auxiliary arcs encode bounds already implied by the cardinality box and do not count toward $|\mathcal{A}|$. Let $\delta_{i,j}$ denote the resulting shortest-path distances. By the same path-summation and feasible-vector argument used above, with $C_k = \delta_{k,j}$, equivalence implies

$$\delta_{i,j} = \max_{\boldsymbol{C} \in \mathcal{F}_0} (C_i - C_j) = d_{i,j}, \qquad i, j \in \mathcal{M}.$$

We first establish a lower bound on the constraints needed within each non-singleton class. For any $Q \in \mathcal{Q}$ with $|Q| \ge 2$, its members must be mutually connected by zero-length paths because $\delta_{i,j} = \delta_{j,i} = 0$ for all $i, j \in Q$. Such paths cannot use auxiliary arcs, whose length is $N > 0$. They also cannot leave Q: any outside node on such a path would have mutual zero distances with a member of Q and hence belong to Q. Therefore, the zero-length arcs within Q form a strongly connected directed graph. Every node requires at least one outgoing arc within Q, so

$$|\mathcal{A} \cap (Q \times Q)| \ge |Q|.$$

We next establish a separate lower bound for each ordered class pair retained in Step 3.2. Recall that E denote the ordered class pairs retained in Step 3.2. Fix $(Q, R) \in E$, $i \in Q$, and $j \in R$. A shortest i-j path has length $d_{Q,R} < N$ and therefore uses no auxiliary arc. If it visited a node k in a third class T, its length would satisfy

$$d_{Q,R} \ge \delta_{i,k} + \delta_{k,j} = d_{Q,T} + d_{T,R} > d_{Q,R},$$

where the strict inequality follows from the retention condition in Step 3.2. This argument also applies when $d_{Q,R} = 0$. Thus, this path stays within $Q \cup R$ and must contain an arc from Q to R. This implies

$$|\mathcal{A} \cap (Q \times R)| \ge 1.$$

Combining these disjoint lower bounds establishes the minimum cardinality of $\mathcal{A}'$. The within-class arc sets and the arc sets associated with distinct ordered class pairs are mutually disjoint, so

$$|\mathcal{A}| \ge \sum_{\substack{Q \in \mathcal{Q} \\ |Q| \ge 2}} |Q| + |E| = |\mathcal{A}'|.$$

The equality holds because Step 3.1 selects exactly $|Q|$ arcs for each non-singleton class, and Step 3.2 selects exactly one arc for each pair in E. Since $(\mathcal{A}, \boldsymbol{x}_\mathcal{A})$ was arbitrary and the returned system is equivalent to $\mathcal{F}_0$, the algorithm attains the minimum cardinality.

We also verify that every retained tolerance level is tight. For each $(i,j) \in \mathcal{A}'$, the algorithm uses the tolerance $d_{i,j}$. By (C.10), the feasible vector $C_k = d_{k,j}$ attains $C_i - C_j = d_{i,j}$. Replacing this tolerance by any smaller nonnegative integer would therefore exclude this feasible vector and change the feasible region. When $d_{i,j} = 0$, no smaller nonnegative integer is admissible.

Finally, we establish the claimed running time. Step 1 and the construction of the equivalence classes and their cycles require $O(M^2)$ time. The Floyd-Warshall computation in Step 2 takes $O(M^3)$ time, and Step 3.2 examines at most $|\mathcal{Q}|^3 \leq M^3$ triples. The total running time is therefore $O(M^3)$. $\square$